

EMBEDDED SENSORS For Critical Applications

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Availability of real-time data and reliable transmission from the sensor to the receiver are crucial for medical, nuclear and autonomous vehicle applications. This article details how sensors meet the requirements of such applications

Sensing technology is evolving at a very fast pace with miniaturisation of micro-processors, embedded systems and other modules, especially micro-electromechanical system (MEMS) sensors. Many embedded sensors have been incorporated into traditional and existing sensor node platforms. These embedded sensor nodes are often physically small and relatively inexpensive computers, each with some set of sensors or actuators.

Embedded sensors are actually part of embedded computers placed in the physical real world that interact with the surrounding environment. With modern hardware advances in sensors and software components, embedded sensors are an active and growing area of embedded computing, and one can witness their expanding applications in many areas.

For example, embedded sensor module MICA2 Mote is a popular wireless measurement system used for enabling low-power, wireless sensor networks. It supports 868/916MHz, 433/315MHz multi-channel transceivers with extended range. It can be interfaced to measure light, temperature, RH, barometric pressure, acceleration/seismic, acoustic, magnetic and other parameters.

Critical applications

Mission-critical applications require safe computer architectures, including the Internet of Things (IoT) with predictable failure behaviour, to avoid financial loss as well as damage to people and the environment caused by

equipment failure. Mission-critical computer architectures used in railways or avionics are complex and include safety-critical characteristics, reliability, error-behaviour modes, safety integrity levels and various other standards. There are also dedicated development, validation, production and qualification processes.

Mission-critical, time-critical and safety-critical applications require fast identification, timely action, reliable notification and warning messages.

The context and levels of criticality may vary from industry to industry but there are some parameters common to all such applications involving sensors. Availability of real-time data and reliable transmission from the sensor to the receiver are important in critical applications. In these applications, data received after the expiration time will not only render something useless but also negatively impact the decision made by users. Good quality, longevity, high sensitivity and reliability are prerequisites for sensors to deal with such critical applications.

This article covers some sensors used in critical applications such as medical, nuclear and autonomous vehicles.

Medical sensors

A medical device is used to diagnose, monitor or treat diseases, or as a supportive aid for physically disabled persons. Various types of sensors are used in medical applications, including temperature probes, force sensors in kidney dialysis machines, airflow sensors in anaesthesia delivery systems, and pressure sensors in infusion pumps and sleep apnea machines. Implanted pacemaker is a real-time embedded sensor system that delivers a synchronised rhythmic electric stimulus to the heart muscle in order to maintain an effective cardiac rhythm.

Medical devices are segmented into different classes. For example, in Canada, Class I devices present the lowest potential risk and Class IV devices the highest potential risk. Implantable pacemakers are high-risk devices because if the device fails, it costs a precious life. Sensors/

Fig. 1: Mica2 Mote
(Courtesy: www.eecs.harvard.edu)

